A node doesn’t need to contain actors other than the `ActorSystem`, so it is possible to use pure seed nodes. Figure 8.3 shows how a first seed node initializes a cluster and how other nodes join the cluster.

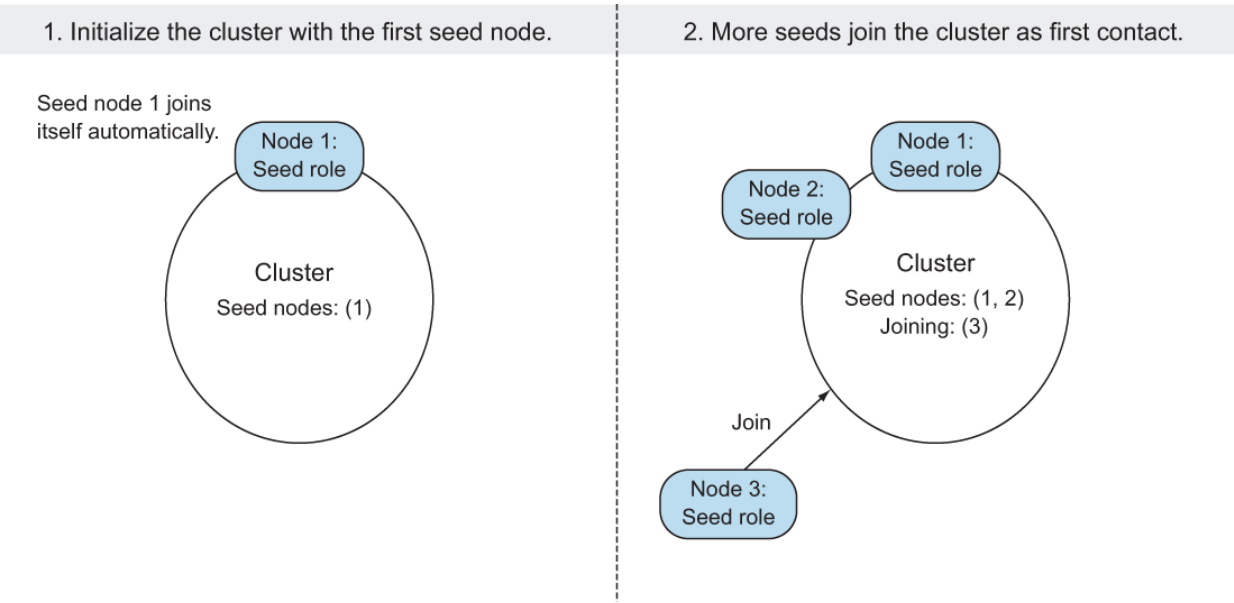


Figure 8.3 Initializing a cluster with seed nodes

Seed nodes can be defined in three ways. Note that these different methods must not be mixed:

- *Configuration*—Passing a configuration of initial node addresses, usually for local development or testing. Using them in production is not recommended.
- *Using Cluster Bootstrap*—Programmed to automatically find nodes in cloud environments such as Kubernetes, AWS, Google Cloud, Azure, and others. This is the recommended approach for use in such environments.